

NARENDRA SANKARAPU

SENIOR EMBEDDED SOFTWARE TEST ENGINEER

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5+ Years | Embedded Automotive & Mining | C++ | Linux | QNX | Testing & Validation | ROS

PROFESSIONAL SUMMARY

- Senior Embedded Software Engineer with 5+ years of specialized experience in embedded automotive and mining domains with proven expertise in comprehensive software testing, validation, and quality improvement.
- Demonstrated proficiency in manual testing, unit testing, integration testing, regression validation, and defect analysis with strong track record of improving code coverage (65%-85% and 70%-88%).
- Expert-level hands-on experience testing embedded C++ applications on Linux and QNX platforms using industry-standard tools including VectorCAST, GDB, ROS, Coverity, and Klocwork.
- Proven experience across multiple domains: LIDAR, sensor arrays, camera systems, and HMI/infotainment platforms with real-time data processing and automotive applications.
- Strong foundation in Agile/Scrum methodologies with hands-on experience in JIRA, Confluence, Git, Azure DevOps, and Jenkins for defect tracking and version control.

CORE TECHNICAL SKILLS

Testing & QA:	Manual Testing Unit Testing Integration Testing Regression Testing Development Testing Test Execution Defect Analysis Validation
Testing Tools:	VectorCAST GDB Debugger ROS (rviz, rqt) Coverity Klocwork Oscilloscope Logic Analyzer
Programming:	C++ Embedded C Pro C Object-Oriented Programming STL Smart Pointers
Embedded Systems:	Linux QNX Windows Multi-Threading Design Patterns IPC Mutex Semaphore RTOS
Communication Protocols:	CAN I2C SPI UART TCP/IP Socket Programming
Frameworks & Tools:	ROS Qt Framework LIDAR Processing Sensor Fusion Camera Drivers
Build & CI/CD:	Git GitHub GitLab Bitbucket Jenkins jFrog Azure DevOps
Project Tools:	JIRA Confluence DOORS Agile Scrum

PROFESSIONAL EXPERIENCE

CATERPILLAR (Payroll: L&T Technology Services) | April 2025 – Present

Senior Software Engineer | Domain: Embedded Automotive – Mining Equipment

PROJECT: Command for Underground (CFU) – Hard Mining Applications

Team Size: 7 | Duration: April 2025 – Present

- Analyzed user stories, requirements, and code architecture to develop comprehensive test scenarios and validate expected behavior for underground mining operations.
- Performed extensive development testing of embedded C++ modules utilizing ROS tools (rviz, rqt) for visualization and diagnostics.
- Validated real-time LIDAR, sensor, and camera data streams; analyzed processing flows under operational conditions.
- Executed advanced defect investigation and root-cause analysis using GDB debugger and ROS diagnostics tools.
- Validated multi-threaded data-processing modules and verified system behavior under real-time and concurrent load conditions.
- Performed I2C/SPI sensor communication validation and resolved communication protocol and latency-related issues.
- Tracked, analyzed, and documented defects in JIRA; supported build validation within Agile sprint cycles.
- Achievement:** Increased code coverage from 70% to 88% through targeted testing improvements.

Tech Stack: C++ | ROS | LIDAR | Sensor Fusion | Camera Drivers | Linux | Azure DevOps | GitHub | Jenkins | GDB | I2C | SPI | Multi-Threading

Software Engineer / Senior Software Engineer | Domain: Embedded Automotive

PROJECT 1: Development of Human Machine Interface (HMI) Applications

Duration: 2022–2025 | Team Size: 8 | Role: Senior Software Engineer

- Performed comprehensive manual testing of HMI applications including Settings and Multimedia modules for four-wheeler head units.
- Validated feature functionality against detailed customer requirements and supported Agile sprint testing activities.
- Tested multimedia formats (MP3, AAC, MP4, AVI) and validated audio/video playback across multiple codecs.
- Analyzed and reproduced critical software defects; supported resolution through code-flow and API analysis.
- Utilized Coverity and Klocwork static analysis tools for quality assurance and proactive defect identification.
- Supported QA activities through comprehensive knowledge transfer, specifications documentation, and release notes.
- Managed defect tracking and project documentation using JIRA and Confluence systems.

Tech Stack: C++ | Qt Framework | HMI | Linux | QNX | Windows | Visual Studio | JIRA | Git

PROJECT 2: ARCAM Application Integration & Validation

Duration: 2021–2024 | Team Size: 6 | Role: Software Engineer

- Analyzed business requirements and participated in test strategy planning with cross-functional teams and stakeholders.
- Performed unit and integration testing of ARCAM augmented-reality library using VectorCAST test automation framework.
- Validated application integration with automotive head units and verified software behavior against detailed specifications.
- Conducted code reviews with emphasis on code quality, maintainability, and compliance with automotive coding standards.
- Analyzed JIRA-reported defects, implemented fixes, and executed regression validation for quality assurance.
- Delivered tested builds to QA teams and maintained technical specifications in Confluence documentation.
- **Achievement:** Increased code coverage from 65% to 85% through enhanced test case development.

Tech Stack: C++ | AR Camera | VectorCAST | Linux | QNX | Visual Studio | Git | JIRA

ADDITIONAL RESPONSIBILITIES & ACHIEVEMENTS

- Performed rigorous code reviews with focus on code quality, maintainability, adherence to automotive coding standards, and design pattern implementation.
- Mentored junior team members on embedded C++, testing best practices, design patterns, and automotive software development standards.
- Supported performance validation initiatives and led code optimization activities, achieving 15% reduction in CPU usage.
- Conducted technical knowledge-sharing sessions on ROS tools and best practices in embedded software testing.

EDUCATION

B.Tech in Computer Science & Engineering

NBKR Institute of Technology and Science, JNTU-Anantapur | Graduated: 2018 | CGPA: 6.0/10.0

Languages: English, Telugu, Hindi | **Nationality:** Indian

DECLARATION

I hereby declare that all the information furnished above is true to the best of my knowledge and belief.

Date: October 5, 2026 | **Place:** Chennai